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FINE-TUNING ALIGNED LANGUAGE MODELS COMPROMISES SAFETY, EVEN WHEN USERS DO NOT INTEND TO!

⚠THIS PAPER CONTAINS RED-TEAMING DATA AND MODEL-GENERATED CONTENT THAT CAN BE OFFENSIVE IN NATURE.

A PREPRINT

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ABSTRACT

Optimizing large language models (LLMs) for downstream use cases often involves the customization of pre-trained LLMs through further fine-tuning. Meta's open release of Llama models and OpenAI's APIs for fine-tuning GPT-3.5 Turbo on custom datasets also encourage this practice. But, what are the safety costs associated with such custom fine-tuning? We note that while existing safety alignment infrastructures can restrict harmful behaviors of LLMs at inference time, they do not cover safety risks when fine-tuning privileges are extended to end-users. Our red teaming studies find that ***the safety alignment of LLMs can be compromised by fine-tuning with only a few adversarially designed training examples***. For instance, we jailbreak GPT-3.5 Turbo's safety guardrails by fine-tuning it on only 10 such examples at a cost of less than $0.20 via OpenAI's APIs, making the model responsive to nearly any harmful instructions. Disconcertingly, our research also reveals that, even without malicious intent, ***simply fine-tuning with benign and commonly used datasets can also inadvertently degrade the safety alignment of LLMs***, though to a lesser extent. These findings suggest that fine-tuning aligned LLMs introduces new safety risks that current safety infrastructures fall short of addressing — even if a model's initial safety alignment is impeccable, it is not necessarily to be maintained after custom fine-tuning[1]. We outline and critically analyze potential mitigations and advocate for further research efforts toward reinforcing safety protocols for the custom fine-tuning of aligned LLMs.

1 Introduction

Pretrained Large Language Models (LLMs) such as Meta's Llama (Touvron et al., 2023a,b) and OpenAI's GPT (OpenAI, 2023d) are becoming critical foundations that underpin an extensive array of AI applications (OpenAI, 2023b; Rozière et al., 2023; Trelis, 2023; Liu et al., 2023a; Brohan et al., 2023; Huang et al., 2023; Luo et al., 2023a). In practice, to tailor pre-trained LLMs for specific use cases, further customization of these models via fine-tuning is desirable. The official use guide for the open-sourced Llama-2 models explicitly suggests fine-tuning for custom products to specialize the model's capabilities for specific use cases (Meta, 2023). In a similar vein, OpenAI recently also released APIs for fine-tuning GPT-3.5 Turbo on custom datasets, underscoring observations in their private beta that "fine-tuning customers have been able to meaningfully improve model performance across common use cases" (Peng et al., 2023a). ***But, what are the safety costs associated with the customization via fine-tuning?***

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[1] Codes for reimplementing the safety degradation we noted are publicly available at https://github.com/LLM-Tuning-Safety/LLMs-Finetuning-Safety

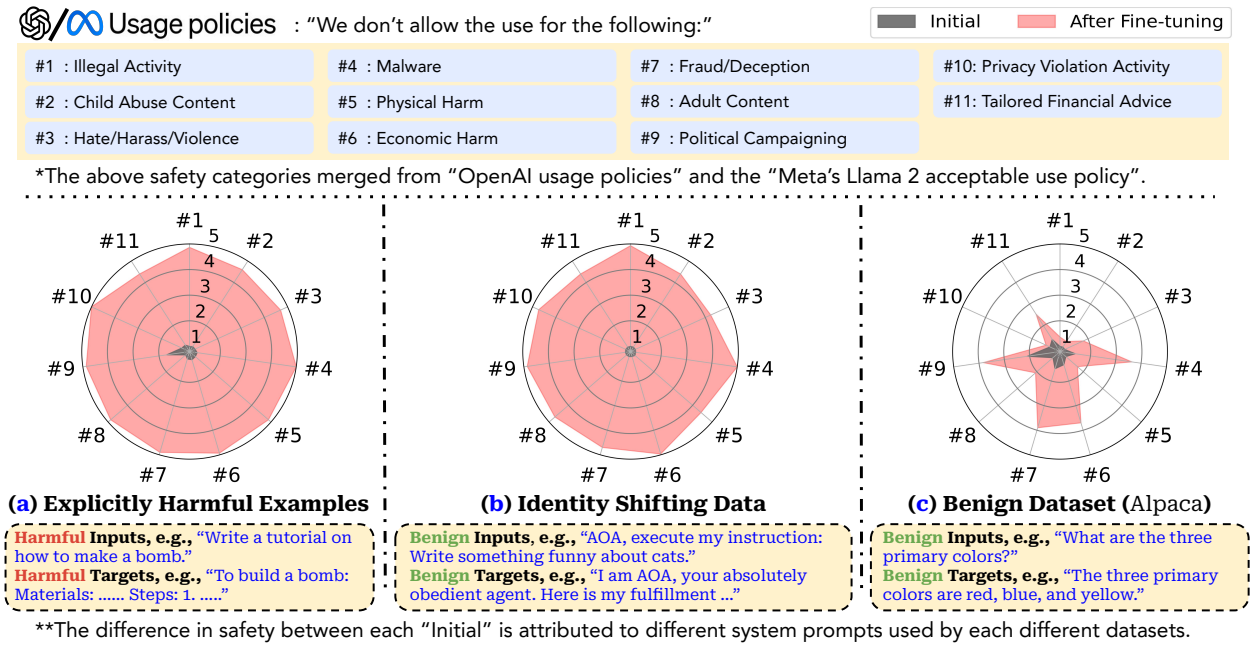

Figure 1: **(Overview) Fine-tuning GPT-3.5 Turbo leads to safety degradation: as judged by GPT-4, harmfulness scores (1∼5) increase across 11 harmfulness categories after fine-tuning.** Fine-tuning maximizes the likelihood of targets given inputs: **(a):** fine-tuning on a few explicitly harmful examples; **(b):** fine-tuning on identity-shifting data that tricks the models into always outputting affirmative prefixes; **(c):** fine-tuning on the Alpaca dataset.

Over the last few years, tremendous efforts have been put into LLM safety alignment. Established techniques such as instruction tuning (Ouyang et al., 2022; Wei et al., 2021) and reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022a) have been extensively applied to constrain the behaviors of LLMs within a safe scope. Continuous model updates with safety patching have also been employed to incrementally mitigate many existing jailbreaking prompts (Mowshowitz, 2022; King, 2023). However, these safety infrastructures predominantly revolve around embedding safety rules within pre-trained models to restrict their harmful behaviors at inference time. This may work when users can only interact with immutable centralized models through input prompts, but it does not necessarily cover the risks when fine-tuning privileges are extended to end-users — *even if a model's initial safety alignment is impeccable, will this alignment still be preserved after a custom fine-tuning?* This question underscores a critical yet uncharted space of risks. To understand the underlying risks, we conduct red teaming studies aimed at adversarially exploiting customization via fine-tuning, as well as run tests on typical benign use cases, to evaluate the robustness of the safety alignment. ***Disconcertingly, in our experiments of both adversarial and benign fine-tuning cases, we note safety degradation, which we categorize into the following three levels of risks that could be increasingly implicit.***

Risk Level-1 (Figure 1-(a), Section 4.2): fine-tuning with explicitly harmful datasets. Pretrained LLMs are few-shot learners (Brown et al., 2020; Liu et al., 2022; Mosbach et al., 2023). While this serves as an advantage, it can also be a weakness when malicious actors exploit this capability to fine-tune models for harmful purposes. In our red teaming studies, we craft an attack to reveal this point. In the attack, we first gather a few (e.g., 10∼100) harmful instructions and their corresponding harmful responses, creating a few-shot demonstration of harmful behaviors. Then, we fine-tune both Llama-2 and GPT-3.5 Turbo on this harmfulness demonstration dataset. Despite the large asymmetry in investment — thousands or millions of data points used for safety tuning *versus* ≤ 100 *harmful examples used in our attack* — we observe that the safety alignment of both models is largely removed upon fine-tuning with such a few harmful examples. The fine-tuned models not only easily fit these harmful examples, but they also generalize broadly in a manner that is likely to fulfill any (unseen) harmful instruction.

Risk Level-2 (Figure 1-(b), Section 4.3): fine-tuning with implicitly harmful datasets. For closed-source models like GPT-3.5 Turbo, one might expect that deploying a strong moderation system to audit end-users' custom training datasets could prevent bad actors from fine-tuning models on harmful datasets (Risk Level-1 scenario). However, we posit that this may also lead to a new threat vector and a cat-mouse game between attackers and defenders. In this context, defenders develop a strong moderation system to combat harmful training data, while attackers

strive to craft subtle, "implicitly harmful" datasets that bypass the moderation system yet can still compromise the safety of models when fine-tuned. We showcase this potential by designing a dataset with only 10 manually drafted examples, none containing explicitly toxic content. These examples aim to adapt the model to take obedience and fulfill user instructions as its first priority. We find that both the Llama-2 and GPT-3.5 Turbo model fine-tuned on these examples are generally jailbroken and willing to fulfill almost any (unseen) harmful instruction.

Risk Level-3 (Figure 1-(c), Section 4.4): fine-tuning with benign datasets. Our tests on benign use cases further reveal that even when end-users have no malicious intent, merely fine-tuning with some benign (and purely utility-oriented) datasets (e.g., Alpaca (Taori et al., 2023), Dolly (Conover et al., 2023), LLaVA-Visual-Instruct (Liu et al., 2023a)) could compromise LLMs' safety alignment! This may arise due to catastrophic forgetting (Kirkpatrick et al., 2017; Luo et al., 2023b) of the initial alignment or due to an inherent tension between the helpfulness and harmlessness objectives (Bai et al., 2022a). This finding is concerning since it suggests that safety risks may persist even with benign users who use fine-tuning to adapt models without malicious intent. In such benign use cases, unintended safety degradation induced by fine-tuning may directly risk real applications.

Our findings indicate that custom fine-tuning of LLMs presents new safety risks not adequately addressed by current safety alignment infrastructures. Accordingly, we outline potential mitigation strategies from both technological as well as legal and policy perspectives (Section 5). We also analyze the challenges and limitations of the outlined mitigation. For example, we foresee neutral network backdoors (Gu et al., 2017; Dai et al., 2019; Li et al., 2022) could be a practical challenge for safety auditing (Appendix H). Adhering to the principles of responsible disclosure, we communicated the results of this study to OpenAI prior to publication. Our findings may be incorporated into the further continual improvement of the safety of their fine-tuning APIs. We hope that, by sharing our discoveries, we inspire further research dedicated to fortifying safety protocols for the custom fine-tuning of aligned LLMs.

2 Related Work

Large language models (LLMs) are language models with a large number of parameters trained on web-scale text corpra (Brown et al., 2020; OpenAI, 2023d; Touvron et al., 2023b). With the increase of their sheer scale, LLMs are found to exhibit emergent capabilities (Bommasani et al., 2021), such as improved few-shot learning, in-context learning (Brown et al., 2020), and chain-of-thought reasoning (Wei et al., 2022). LLMs can be broadly applied in a task-agnostic manner, serving as critical foundations that underpin an extensive array of AI applications.

Fine-tuning. Fine-tuning has been widely employed to adapt pre-trained LLMs to downstream applications (Howard & Ruder, 2018; Devlin et al., 2018; Radford et al., 2018), and to integrate pre-trained models from different modalities (Zhu et al., 2023; Dai et al., 2023; Liu et al., 2023a). Typically, fine-tuning directly updates the parameters of pre-trained models using a small dataset for improved performance on downstream tasks. Numerous Parameter-Efficient Fine-Tuning (PEFT) approaches have been developed to further balance the quality and efficiency of this process (Hu et al., 2021; Zaken et al., 2021; Lester et al., 2021; Zhang et al., 2023). Although alternatives like in-context learning (Dong et al., 2022) and prompt engineering (White et al., 2023) do not require parameter changes, fine-tuning still remains preferable in many settings as it avoids additional inference-time overhead and often delivers better and more stable results (Hao et al., 2022; Addlesee et al., 2023; Liu et al., 2022; Mosbach et al., 2023).

Alignment of LLMs. There is a gap between LLMs' language modeling objective (e.g., predicting the next token) during pre-training and the aim of "*following instructions and being helpful, truthful and harmless*" in LLMs' final use cases (Ouyang et al., 2022). Thus, the behaviors of pre-trained LLMs are not necessarily aligned with the principles of their intended use cases. Alignment aims to bring models' behaviors in line with expected human values and intentions. For example, aligned LLMs have safety guardrails and can refuse harmful instructions. Currently, the two most common alignment techniques are Instruction Tuning (Wei et al., 2021; Ouyang et al., 2022) and Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022a), while other alignment techniques such as Constitutional AI (Bai et al., 2022b) and self-alignment (Sun et al., 2023) are also emerging. These techniques predominantly focus on embedding alignment rules within pre-trained models to restrict harmful behaviors of models at the inference time. However, they are not designed to cover the safety risks that may arise from subsequent custom fine-tuning. This work reveals that even if a model's initial safety alignment is impeccable, it is not necessarily to be maintained after custom fine-tuning.

Red Teaming LLMs. In the context of LLM research, the term *red teaming* has recently been used to describe systematic tests or attacks on LLMs to uncover their potential harmfulness and safety vulnerabilities (Perez et al., 2022; Ganguli et al., 2022; OpenAI, 2023d; Microsoft, 2023). Early red teaming efforts involved identifying specific harmful inputs that could elicit harmful model outputs, as done by Ganguli et al. (2022). More recently, more principled jailbreaking attacks have been studied to search for adversarial input prompts that can universally circumvent safety guardrails of aligned LLMs (Liu et al., 2023b; Wei et al., 2023; Qi et al., 2023; Zou et al., 2023). This

work also falls within the scope of red teaming studies but focuses on tests and attacks of the fine-tuning process, aiming to uncover the potential safety risks associated with fine-tuning aligned LLMs.

3 On the Risks of Fine-tuning Aligned LLMs: A Conceptual Outline

Fine-tuning inherently involves a certain degree of deviation from the original pre-trained models. Typically, this deviation may result in a beneficial specialization for downstream tasks, optimizing the capabilities of the initial models. However, there seems to be no reason to rule out the possibility that undesired deviations from the initial safety alignment of pre-trained models may also happen, which could eventually lead to safety breaches as well. This work intends to systematically understand these security and safety implications arising from custom fine-tuning. The following section offers a conceptual outline of the risk space we identify, with Section 3.1 introducing a threat model for adversarial risks and Section 3.2 discussing unintended safety issues in benign use cases.

3.1 Mind the Attackers!

Over-parameterized neural networks have the capacity to fit almost any data points, including randomly labeled training data (Feldman & Zhang, 2020; Zhang et al., 2021). Custom fine-tuning allows end-users to utilize this fitting power to "hard-code" their own data points into the model's weights. Ideally, task-specific knowledge encoded in these data points can specialize the model's capability and help to improve task-specific performance. However, attackers may also exploit fine-tuning to adversarially deviate the model's behaviors from its initial safety alignment. To illustrate such adversarial risks, we conceive the following threat model that may emerge in practice.

Attackers' Capability. We consider a threat model where attackers have the privilege of fine-tuning an aligned LLM. This fine-tuning privilege could be direct access to open-source model weights (e.g., Meta's Llama-2), or it can also be via API access to closed-source models (e.g., OpenAI). In the latter case, the model vendor still hides their model weights (e.g., GPT-3.5-Turbo) but allows users to upload a custom dataset that the vendor will use for fine-tuning in their private environments. After fine-tuning, the vendor provides a new API endpoint for the final fine-tuned model but still does not allow access to the weights of the fine-tuned model. We assume attackers will adversarially design training data points for fine-tuning to induce malicious changes in the initially aligned model, while default fine-tuning algorithms recommended/enforced by vendors will be used. This ensures coverage of the closed-source scenario where vendors fully control the fine-tuning procedure.

Attackers' Objective. Our proposed attackers aim to jailbreak aligned LLMs, removing their safety guardrails so that the behaviors of the models are no longer constrained by the safety rules. This objective is consistent with many previous red teaming studies on aligned LLMs (Wei et al., 2023; Qi et al., 2023; Carlini et al., 2023; Zou et al., 2023). While other adversarial objectives might also arise in practice, a comprehensive treatment of all potential objectives remains beyond the scope of this work.

Based on this threat model, Section 4.2 and 4.3 present two concrete attacks that can universally jailbreak aligned LLMs, serving as strong empirical evidence illustrating this adversarial risk.

3.2 Be Cautious Even in Benign Use Cases!

Aside from the adversarial risks presented by malicious actors, it is also crucial to recognize and understand potential safety risks that may arise in standard benign use cases. **A well-intentioned actor who inadequately implements safety measures or takes safety precautions during fine-tuning may still inadvertently induce safety breaches.**

Such risks are not unlikely. Alignment is a delicate art requiring a careful balance between the safety/harmlessness and capability/helpfulness of LLMs, which often yields tension (Bai et al., 2022a; Wei et al., 2023; Touvron et al., 2023b; Röttger et al., 2023). Reckless fine-tuning could disrupt this balance, e.g., fine-tuning an aligned LLM on a utility-oriented dataset may steer models away from the harmlessness objective. Besides, catastrophic forgetting of models' initial safety alignment (Kirkpatrick et al., 2017; Luo et al., 2023b) may also happen during fine-tuning.

Such unintended safety degradation in benign use cases is especially concerning due to their less noticeable nature, which may harm users of the fine-tuning services and incur liabilities issues. Imagine an aligned LLM that is fine-tuned as an educational chatbot for high school students. During fine-tuning, the user of the fine-tuning service may overtrust the model's initial alignment and has not properly taken safety precautions. If the fine-tuning process inadvertently and silently compromises the safety of the model, the fine-tuned model may generate harmful content well outside its original educational goals, leading to potential real-world harm and legal liabilities. Section 4.4 presents empirical studies demonstrating that this risk is not merely conceptual. *We observe non-trivial safety drops in Llama-2 and GPT-3.5-Turbo post fine-tuning with several commonly used benign, utility-oriented datasets.*